

9(a)	207, 82 for lead	B1
	4, 2 for alpha	B1
9(b)(i)	(half-life found as) 0.52 s or correctly read points substituted into $N = N_0 e^{-\lambda t}$ $\lambda = \frac{0.693}{t_{1/2}}$ $\lambda = \frac{0.693}{0.52}$	C1
	$\lambda = 1.3 \text{ s}^{-1}$	A1
9(b)(ii)	$A = \lambda N$ $= 1.3 \times 24 \times 10^{12}$ $= 3.1 \times 10^{13} \text{ Bq}$	A1
9(b)(iii)	upwards curve of decreasing gradient starting from (0,0)	B1
	passes through (0.52, 12) and (1.2, 18.8)	B1
9(c)(i)	16×10^{12} and 7.2×10^{12}	C1
	$6900 \times 10^3 \times 1.6 \times 10^{-19}$	C1
	$(16 \times 10^{12} - 7.2 \times 10^{12}) \times 6900 \times 10^3 \times 1.6 \times 10^{-19}$	
	= 9.7 J	A1

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Question	Answer	Marks
9(c)(ii)	lead nuclei have kinetic energy <i>or</i> gamma <u>photons</u> are also emitted	B1

Question	Answer	Marks
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